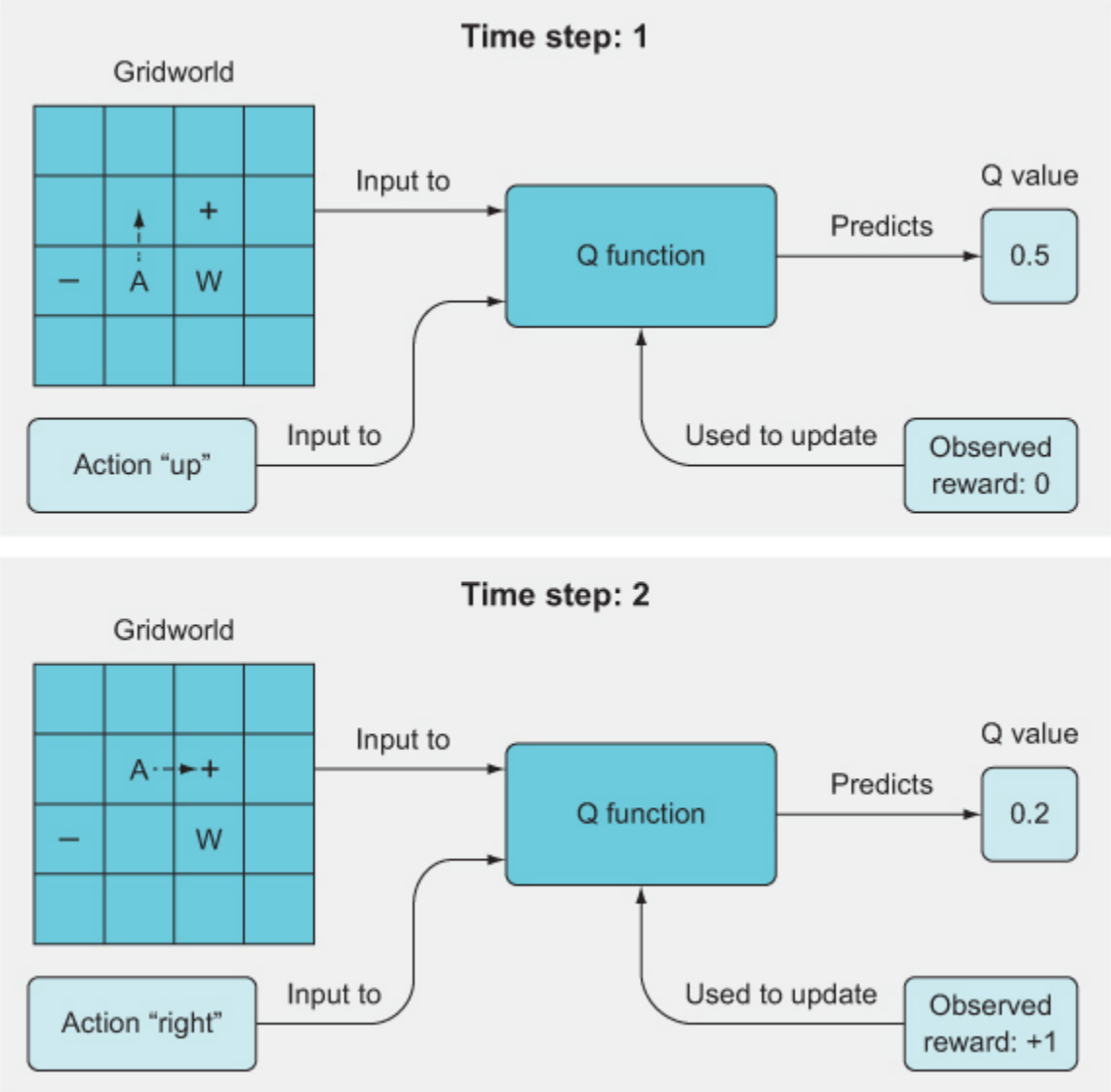
negative reward or a really big reward, and we want to improve the accuracy of the algorithm's predictions (see figure 3.3).

Figure 3.3. Schematic of Q-learning with Gridworld. The Q function accepts a state and an action, and returns the predicted reward (value) of that state-action pair. After taking the action, we observe the reward, and using the update formula, we use this observation to update the Q function so it makes better predictions.



4. Now we run the algorithm using S_{t+1} as input and figure out which action our algorithm predicts has the highest value. We'll call this value $Q(S_{t+1},a)$. To be clear, this is a single value that reflects the highest predicted Q value, given our new state and all possible actions.
5. Now we have all the pieces we need to update the algorithm's parameters. We'll perform one iteration of training using some loss